Peter Samaan

Atlanta, GA | (678) 938-7787 | psamaan3@gatech.edu | U.S. Citizen | linkedin.com/in/peter-samaan

Education

Georgia Institute of Technology | Atlanta, GA *January 2025 – Present*

Bachelor of Science in Electrical Engineering, Cumulative GPA: 3.39 Expected Graduation, May 2027

- Relevant Coursework: Signals and Systems, Intro to Signal Processing, Digital Design Lab, Advanced Programming Techniques for Engineering Applications, Engineering Software Design, Electromagnetics, Microelectronic Circuits

Kennesaw State University | Marietta, GA *August 2023 – December 2024*

Transfer with 69 Credit Hours, GPA: 3.72

- Relevant Coursework: Circuit Analysis, Digital Logic Design, Intro to Programming (C++), Probability and Data Analysis

Skills

Embedded & Hardware: Microcontroller Development (Arduino, ARM mbed), FPGA Development (VHDL), Digital Logic Design, Oscilloscope Measurement, Logic Analyzer Measurement, Hardware-Software Integration, Soldering
Signal Processing: Digital Signal Processing (DSP), Digital Filter Design, Analog Filter Design, FFT Spectral Analysis
Programming: C/C++, Python (NumPy, SciPy), MATLAB, VHDL, Assembly, R, Multithreading, OpenMP
Engineering Software: Quartus Prime, Circuit Simulation (LTspice, Multisim), AutoCAD Electrical, MS Office

Experience

Diversified Automation Inc. | Louisville, KY *May – July 2025*

Hardware Engineering Co-Op

- Edited and updated technical hardware drawings for 5+ client-based contracts using AutoCAD, adhering to project specifications and client needs.
- Revised 200+ schematics by applying a detail-oriented approach to ensure compliance with various formats.
- Collaborated with engineers to revise layout designs, contributing to the accuracy and efficiency of project documentation.

Projects

ECG Signal Processing Pipeline *March 2026*

- Analyzed and processed recorded ECG data from the MIT-BIH Arrythmia Database using Python (NumPy, SciPy).
- Designed digital IIR filters (Butterworth bandpass and 60 Hz notch) to remove baseline drift and powerline interference.
- Validated filtering performance in frequency domain using FFT spectral analysis.
- Developed R-peak detection algorithm to identify heartbeats and compute heart rate.

Conway's Game of Life *October 2025*

- Implemented "Conway's Game of Life" in C++ using sequential and parallel processing methods.
- Improved processing speed by 120% using both standard library threads and OpenMP threads.
- Utilized SFML (Simple Fast Media Library) to develop 2D graphics for individual cells.
- Added command line arguments to select processing type, window dimensions, and number of threads.

FPGA LED Brightness Controller *March – April 2025*

- Developed a peripheral device in four-person team using an FPGA and Quartus Prime to interact with external LEDs while maintaining usable interface.
- Designed peripheral device in VHDL with 128-bit brightness resolution.
- Built software demonstration using assembly to directly showcase functionality by rapidly changing brightness values across multiple LEDs on the board simultaneously.
- Implemented gamma-correction LUT in VHDL to improve accuracy of LED brightness values.

Traffic Light Design Project *October – November 2024*

- Designed an 8-state state machine with 5 inputs and 6 outputs, as well as a car counter system for tracking the number of vehicles waiting.
- Constructed circuit for I/O including 3 car counters, 3 CMOS timers, a dip switch indicating an incoming train, and LEDs.

Leadership

Orthodox Chrisitan Campus Ministries (OCCM) | Atlanta, GA *August 2025 – Present*

Vice President

- Organize and plan weekly edifying events for 50+ undergraduate and graduate students from 3 universities.
- Oversee monthly community service and outreach initiatives both on and off campus.

St. Mark's American Coptic Orthodox Church | Doraville, GA

Youth Group Leader *August 2023 – Present*

- Engage class of 15+ students with activities, lessons, and service events to emphasize Orthodox Christian teachings and values.
- Foster connections between volunteers and students via outings and visitations.

Worship Team Leader *November 2022 – Present*

- Engage congregation of 100+ with both contemporary and traditional worship music.
- Plan and lead monthly worship nights to encourage communal worship and prayer.